

OpenSG: A FEniCSx-Based Implementation of the Mechanics of Structure Gene with Emphasis on Aperiodic Beams

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Summary

OpenSG is an open-source platform for multiscale structural mechanics built with a FEniCSx (Baratta et al., 2023) backend. It implements the Mechanics of Structure Genome (MSG) theory (Wenbin Yu, 2019), a unified and rigorous framework for modeling advanced structures with general anisotropy and heterogeneity, including beams, plates, shells, and continuum structures.

OpenSG is a micromechanics code, which means it enables users to efficiently incorporate small-scale details into the global structural scale through constitutive modeling. This involves two primary steps, homogenization and dehomogenization. Homogenization is the process of obtaining homogenized structural properties whereas, dehomogenization is recovering the local stress and displacement fields. These operations are performed over a user-defined domain called the Structure Gene (SG). An SG can be 1-dimensional (line elements), 2-dimensional (quadrilaterals or triangular elements), or 3-dimensional (hexahedron or tetrahedron elements). Using SGs, OpenSG produces structural properties in terms of Cauchy continuum model, plate/shell model, or beam model. Conventional finite element analysis tools can utilize these properties as inputs to predict global structural responses. OpenSG can use these responses to compute small-scale stresses and deformations throughout the SG. In this way, OpenSG provides an accurate, fast, and versatile platform for analyzing structural profiles with key small-scale details like aircraft, wind turbine blades, and additively manufactured parts.

Statement of need

No existing open-source tool provides MSG theory for Timoshenko beam modeling of aperiodic solid SGs, and shell cross-sections with local buckling predictions. OpenSG fills this gap with an open-source MSG implementation Bagla et al. (2023) that supports aperiodic beam SGs composed of shell (Bagla et al., 2026, 2025a) or solid (Bagla et al., 2026) elements. This is especially relevant for nonprismatic structures such as wind turbine blades, tapered rods, propellers, and aircraft wings where 3D phenomena like local panel buckling cannot be captured by beam-only approaches.

State of the Field

Several tools exist for computing structural properties of composite beams and blades. SwiftComp (Wenbin Yu, 2016) is the primary reference implementation of MSG theory, unifying VAMUCH (Wenbin Yu & Tang, 2007) for material unit cells, VAPAS (Wenbin Yu,

39 2005) for plates and shells, and VABS (W. Yu et al., 2002) for beams. However, SwiftComp is
40 commercial and closed-source, limiting reproducibility and community extensibility. BECAS
41 (Blasques, 2012), ANBA4 (Morandini et al., 2010), and PreComp (Bir, 2005) are open-source
42 beam cross-section analysis tools widely used in wind energy, but they are restricted to beam
43 models and do not implement the MSG framework.

44 Software Design

45 OpenSG is built on FEniCSx (Baratta et al., 2023), a modern finite element framework whose
46 Unified Form Language (UFL) allows the MSG governing equations to be expressed in a
47 form that closely mirrors their mathematical derivations. This keeps the implementation
48 auditable and accessible to researchers working directly from the MSG literature. FEniCSx's
49 comprehensive Python API and active development community make it a natural foundation
50 for an extensible, open-source structural analysis tool.

51 The codebase is organized around element formulation type: shell and solid element formulations
52 are implemented in `opensg.core.shell` and `opensg.core.solid` respectively, with shared
53 utilities for material property assignment and stiffness matrix assembly in `opensg.utils`.
54 Material properties and composite layup definitions are supplied via YAML input files and
55 parsed through `opensg.io`. Mesh generation is managed through `opensg.mesh`, which interfaces
56 with external tools such as Gmsh to prepare the necessary data structures to apply the MSG
57 algorithms.

58 Research Impact Statement

59 The development team has applied OpenSG to advance structural analysis of composite
60 structures, including 3D shell-to-beam modeling (Bagla et al., 2026) and wind turbine blade
61 modeling (Bagla et al., 2025b). These applications highlight the potential of OpenSG to enable
62 accurate, scalable, and open-source solutions for structural modeling challenges across wind
63 energy and beyond.

64 The repository includes detailed documentation and multiple Python notebook examples to
65 support external adoption. The test suite includes regression tests for shell and solid stiffness
66 computations against stored baseline results, ensuring reproducibility and providing a clear
67 record of how outputs evolve as the source code is updated. OpenSG is part of the WETO
68 Software Stack, a collection of integrated modeling tools for wind energy research; integration
69 with other WETO codes allows researchers to incorporate OpenSG directly into existing wind
70 energy workflows without requiring standalone preprocessing pipelines.

71 AI usage disclosure

72 No generative AI tools were used in the development of this software or the preparation of this
73 manuscript.

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76 and Purdue University within the Holistic, Multi-Fidelity Wind Farm Design Optimization and
77 Model Coordination project under the U.S. Department of Energy Wind Energy Technologies
78 Office. It is currently being evaluated for its suitability to transform wind turbine blade design,
79 mainly due to its ability to incorporate high-fidelity structural details into efficient beam models.

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